

Functional Annotation of *panC* (PP_4700, UniProt Q88DW8) in *Pseudomonas putida* KT2440

1. Summary

panC (ordered locus **PP_4700**; UniProt **Q88DW8**) of *Pseudomonas putida* KT2440 encodes **pantothenate synthetase** (also called pantoate— β -alanine ligase or pantoate-activating enzyme; EC 6.3.2.1; KEGG ortholog **K01918**). Its primary function is to catalyze the final, committed step of *de novo* **pantothenate (vitamin B5) biosynthesis**: the **ATP-dependent condensation of (R)-pantoate and β -alanine to form pantothenate**, proceeding through a covalent **pantoyl-adenylate** reaction intermediate and releasing **AMP and pyrophosphate**. Pantothenate is the obligate precursor of the 4'-phosphopantetheine moiety of **coenzyme A (CoA) and acyl carrier protein (ACP)**, making PanC an essential feeder of central CoA/ACP-dependent metabolism. The enzyme functions as a **soluble, cytoplasmic homodimer** requiring a **divalent metal cofactor (Mg^{2+} or Mn^{2+})**. The gene is co-organized with *panB* in an overlapping **panB–panC operon**, coupling two consecutive pathway steps.

No experimental study has been published on the *P. putida* KT2440 protein specifically; the annotation is transferred with high confidence from closely related, biochemically and structurally characterized orthologs (*E. coli*, *M. tuberculosis*, plants, fungi) supported by ~53–55% sequence identity, an intact catalytic motif, conserved substrate-binding residues, and native operon context.

2. Gene / Protein Identity Verification

The gene symbol **panC** matches the protein description in every respect and is **not ambiguous** in this context:

- **UniProt Q88DW8** annotates the protein as *Pantothenate synthetase* (EC 6.3.2.1), gene *panC*, locus PP_4700, in *Pseudomonas putida* KT2440.

- **KEGG** independently curates PP_4700 as *panC* / K01918 / EC 6.3.2.1, in the *Pantothenate and CoA biosynthesis* (ppu00770) and *β -Alanine metabolism* (ppu00410) pathways (NCBI protein AAN70273).
- The protein carries the family-defining **Pantoate_ligase** Pfam domain (PF02569; InterPro IPR003721), a C-terminal Pantoate_ligase_C domain (IPR042176), and a Rossmann-like a/b/a fold (IPR014729) — all consistent with the pantothenate synthetase family.
- All literature discussed below concerns the **same enzyme (pantothenate synthetase / PanC / EC 6.3.2.1)** across bacteria, fungi, and plants; the mechanism and reaction are conserved across the family, so cross-species inference is valid.

Conclusion: the target is correctly identified as bacterial pantothenate synthetase.

3. Primary Function: Reaction Catalyzed and Mechanism

3.1 Overall reaction

PanC catalyzes:



This is the "ATP-dependent condensation of pantoate and beta-alanine to form pantothenate" (Wang & Eisenberg, 2006,).

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3.2 Two-step (ordered) mechanism via a pantoyl-adenylate intermediate

Crystallographic "snapshots" of the *M. tuberculosis* ortholog captured the enzyme bound to substrates (ATP/AMPCPP, pantoate), the covalent reaction intermediate, and product (AMP), establishing an ordered, two-step mechanism (Wang & Eisenberg, 2006,).

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1. **Adenylation:** pantoate attacks the α -phosphate of ATP, forming enzyme-bound **pantoyl-adenylate** + PPi.

2. **Aminolysis:** the amino group of β -alanine attacks the activated pantoyl carbonyl, forming the amide bond of pantothenate and releasing AMP.

A key mechanistic feature is that **β -alanine can bind only after the pantoyl-adenylate intermediate has formed** — "binding of beta-alanine in the active site cavity can occur only after formation of the pantoyl adenylate intermediate," with "the phosphate group of AMP [serving] as an anchor for the binding of beta-alanine" (Wang & Eisenberg, 2006,).

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This adenylate-forming chemistry (activation of a carboxylate as an acyl-adenylate) is characteristic of the pantothenate synthetase family and is echoed by the ATP-sulfurylase-like nucleotide-binding motif noted in the KEGG annotation.

3.3 Conserved catalytic residues

Structural/mutational analysis of the *M. tuberculosis* enzyme identifies conserved active-site residues **His44, His47, Asn69, Gln72, Lys160, Gln164** as important for substrate binding and pantoyl-adenylate formation; alanine substitutions (e.g., H47A, K160A) weaken ATP binding and reduce catalysis (Pandey et al., 2018,).

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In *P. putida* Q88DW8 the corresponding catalytic **HxGH motif is intact (His34-Ser35-Gly36-His37)**, and UniProt annotates **His37 as the catalytic proton donor**, along with ATP-binding residues (30–37, 149–152, 178, 186–189), (R)-pantoate-binding residues (Gln61, Gln155), and a β -alanine-binding residue (Gln61) — matching the two-substrate/ATP active-site architecture.

4. Substrate Specificity and Cofactor

- **Strict substrate requirement:** the purified enzyme "requires d-pantoate, beta-alanine and ATP for activity" (Genschel et al., 1999,).

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- **Stereo/chemo-selectivity:** "Neither oxopantoate nor pantoyl-lactone can replace pantoate as substrate," demonstrating tight specificity for reduced, open-chain D-(R)-pantoate (Genschel et al., 1999,).

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- **Relative affinities:** the enzyme "has a higher affinity for pantoate (K_m 45 μ M) than for beta-alanine (K_m 990 μ M)" (Genschel et al., 1999,

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with $k_{cat} \approx 0.63 \text{ s}^{-1}$ and uncompetitive substrate inhibition at pantoate $> \sim 1 \text{ mM}$.

- **Metal cofactor:** the enzyme "is most active in the presence of magnesium or manganese"

(

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) consistent with Mg·ATP being the true substrate for adenylate formation.

These kinetic parameters derive from characterized orthologs; given ~53–55% identity and full active-site conservation, they are expected to hold, at least qualitatively, for *P. putida* PanC.

5. Biological Process / Pathway Context

5.1 Final step of the four-enzyme pantothenate pathway

Bacterial *de novo* pantothenate biosynthesis "compris[es] four enzymic reactions catalysed by ketopantoate hydroxymethyltransferase (KPHMT), L-aspartate-alpha-decarboxylase (ADC), pantothenate synthetase (PS) and ketopantoate reductase (KPR) encoded by **panB**, **panD**, **panC** and **panE** genes" (Chakaulya et al., 2008,

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) PanC performs the terminal condensation. Its two substrates are supplied by the other pathway branches: - **(R)-pantoate** branch: α -ketoisovalerate (from branched-chain amino acid metabolism) $\rightarrow$ ketopantoate (**PanB**, EC 2.1.2.11) $\rightarrow$ pantoate (**PanE**). - **β -alanine** branch: L-aspartate $\rightarrow$ β -alanine (**PanD**, aspartate-1-decarboxylase).

Genetics confirm the gene–function assignment: in *E. coli*/*S. typhimurium*, "the **panC** mutants lack pantothenate synthetase," each showing a single enzyme defect (Cronan et al., 1982,

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5.2 Downstream role: essential precursor of CoA and ACP

"Pantothenate (vitamin B5) is the precursor of the 4'-phosphopantetheine moiety of **coenzyme A and acyl-carrier protein**" (Chakauya et al., 2008, [P 18726075](#)).

Pantothenate "is required for the synthesis of coenzyme A (CoA)" (Riske et al., 2023, [P 37762222](#)).

Because CoA and holo-ACP are indispensable for fatty-acid synthesis, the TCA cycle, and myriad acyl-transfer reactions, the pantothenate/CoA pathway is essential and a validated antimicrobial drug target (Wang & Eisenberg, 2006, [P 16460002](#));

Pepi et al., 2022, [P 35015509](#)).

This is especially relevant for the metabolically versatile, solvent-tolerant lifestyle of *P. putida*, which relies heavily on CoA/acyl-CoA metabolism.

5.3 Genomic organization — panB-panC operon

In *P. putida* KT2440, KEGG genome coordinates place **panB (PP_4699)** at 5,340,894–5,341,694 and **panC (PP_4700)** at 5,341,691–5,342,554 on the same strand; the two ORFs **overlap by 4 bp**, the hallmark of translational coupling within a single operon. This mirrors the *E. coli* cluster where "panB and panC lie adjacent to one another" (Merkel & Nichols, 1996, [P 8837478](#)).

The overlap implies coordinated, stoichiometric expression of two consecutive pathway enzymes.

6. Localization

Pantothenate synthetase is a **soluble, cytoplasmic** enzyme. Characterized plant orthologs localize to the cytosol with "no N-terminal extension" ("The Lotus and rice PSs are therefore likely to be cytosolic"; Genschel et al., 1999,

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bacterial PanC likewise has no signal peptide or membrane-spanning region (Q88DW8 is a 287-aa soluble protein). The enzyme is active as a **homodimer** ("Recombinant PS from Lotus ... is a dimer,"

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the *M. tuberculosis* enzyme is also dimeric). Thus *P. putida* PanC carries out its reaction in the **cytoplasm**, where its substrates (pantoate, β -alanine, Mg·ATP) and downstream CoA-biosynthetic enzymes reside.

7. Evidence Summary

Claim	Evidence type	Key source(s)
Catalyzes ATP-dependent pantoate + β -alanine $\rightarrow$ pantothenate	Experimental (enzymology, crystallography of orthologs)	P 16460002 P 10417331
Two-step mechanism via pantoyl-adenylate; ordered substrate binding	Experimental (crystal structures)	P 16460002
Conserved catalytic residues (His pair / HxGH, Lys, Gln, Asn)	Experimental + bioinformatic	P 29343701 ; UniProt Q88DW8
Strict specificity for D-pantoate; Km values	Experimental (kinetics)	P 10417331
Mg ²⁺ /Mn ²⁺ cofactor	Experimental	P 37578562
Final of four steps in pantothenate pathway	Review + genetics	P 18726075 P 7037743
Product is CoA/ACP precursor; pathway essential/drug target	Review	P 18726075 P 37762222 P 35015509
Cytoplasmic, homodimeric	Experimental (orthologs) + sequence	P 10417331
Q88DW8 is a true PanC ortholog (~53–55% id, intact motif)	Bioinformatic (this work)	UniProt; alignment vs P31663, P9WIL5
panB–panC operon in KT2440 (4-bp overlap)	Genomic (this work)	KEGG ppu; P 8837478
High-confidence AlphaFold fold (mean pLDDT 95.7; 92.7% >90)	Structure prediction (this work)	AlphaFold DB AF-Q88DW8-F1

8. Supported and Refuted Hypotheses

- **Supported:** PP_4700 encodes a functional pantothenate synthetase (EC 6.3.2.1) catalyzing the terminal step of pantothenate biosynthesis via a pantoyl-adenylate intermediate; it is cytoplasmic, homodimeric, Mg/Mn-dependent, strictly D-pantoate-specific, and co-expressed with panB.
 - **Refuted / ruled out:** the gene symbol is **not** ambiguous here — no competing gene with the same symbol and a different function fits the sequence, domains, or genomic context. Oxopantoate and pantoyl-lactone are **not** substrates (ruled out experimentally in the ortholog).
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9. Limitations and Future Directions

- **No direct experimental characterization of the *P. putida* KT2440 protein** exists; all enzymology/structure is from orthologs. Function is inferred (with high confidence) from ~53–55% identity, an intact catalytic HxGH motif, conserved substrate-binding residues, and conserved genomic operon context.
 - Kinetic constants (Km, kcat) are ortholog-derived and may differ quantitatively in *P. putida*.
 - **Future work:** recombinant expression and steady-state kinetics of Q88DW8; determination of a *P. putida* PanC crystal or cryo-EM/AlphaFold-validated structure; targeted *panC* deletion/complementation to confirm essentiality and pantothenate auxotrophy in KT2440; transcriptional mapping to confirm the panB–panC operon experimentally.
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Report generated from literature (PubMed) and database (UniProt, KEGG) evidence, plus computational sequence/genomic analysis. All quantitative claims are cited to primary or review literature.
